CS383 · SOFTWARE ENGINEERING

Midterm 472 Solutions

20 marks

Muharib

01 Solution Q1: 1 to 5

#	Answer	Why	Lecture
1	**B**	Estimating cost, scheduling and monitoring the project are the manager's duties.	6
2	**B**	A successful project is on time, within budget and meets the requirements.	6
3	**B**	The plan holds the risks, the resources and the work breakdown.	6
4	**C**	A self-contained task is easier to estimate, manage and test.	6
5	**B**	A milestone is a significant point with a formal output, such as signing off the specification.	6

02 Solution Q1: 6 to 10

#	Answer	Why	Lecture
6	**B**	Identify risks, assess their impact and plan for them before they happen.	6
7	**A**	Selecting the team and assigning roles is part of people management.	6
8	**B**	A safety-critical system with stable requirements suits a plan-driven process.	3
9	**B**	Fewer dependencies allow parallel work and limit the effect of delays.	6
10	**B**	These are the project planning activities listed in the slides.	6

03 Solution Q1: 11 to 16

#	Answer	Why	Lecture
11	**B**	An FR is a service or behaviour the system provides.	7
12	**C**	Response time is a quality attribute (performance), so it is an NFR.	7
13	**B**	User requirements are high-level and readable by non-technical people.	7
14	**B**	"user-friendly and fast" cannot be measured, so it is ambiguous.	7
15	**B**	The five checks: validity, consistency, completeness, realism, verifiability.	8
16	**B**	Constant change and continuous feedback suit Agile.	4

Note

10 of the 16 questions come from Lecture 6 (project management). Every answer is B except question 4 (C) and question 7 (A).

04 Solution Q2: User / System and FR / NFR

Requirement	User / System	FR / NFR
The system should be easy for new students to learn within one lab session.	**User**	**NFR**
The system shall allow an instructor to upload course files in PDF format.	**System**	**FR**
The application shall support at least 500 concurrent users during registration.	**System**	**NFR**
The system shall encrypt all stored passwords using the SHA-256 algorithm.	**System**	**NFR**

- First: ease of learning is a quality attribute (usability), and the statement is general and in the student's language, so it is User and NFR.
- Second: a service the system provides (upload), with a technical detail (PDF format), so it is System and FR.
- Third: a measurable number (500 users) about performance, so it is System and NFR.
- Fourth: a specific algorithm (SHA-256) about security, so it is System and NFR.

Watch out

The word shall does not make a statement an FR. Security, performance and usability are all NFRs even when written with shall.

05 Solution Q3: model choice and calculations

A. Process model

The answer: Agile. The reason: feedback is weekly, features are re-prioritized often, and the customer wants a working version quickly.

B. Sprint velocity

```
Velocity = (18 + 22 + 20 + 24) ÷ 4 = 84 ÷ 4 = 21 story points
```

C. Task scheduling

```
(i)  Sequential:  5 + 8 + 4 + 7 = 24 days
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```
(ii) Task 2 ‖ Task 3  →  max(8, 4) = 8
```

```
     Total = 5 + 8 + 7 = 20 days
```

When two tasks run in parallel, we take only the longer one, because the other finishes before it.